Experimental Physics & Chemistry / Physique-Chimie

Chapter 1: Classical Newtonian Mechanics

Fundamental laws of kinematics: displacement, velocity, and acceleration vectors.

Newton second law of motion: sum of forces equals mass times acceleration.

Analysis of forces acting on an inclined plane with friction.

LEARNING OBJECTIVES & KEY CONCEPTS

Mathematical formulation: Vector calculus applied to point particles.

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Chapter 2: Energy Conservation and Work

Kinetic energy and potential gravitational energy in uniform gravitational fields.

Theorem of mechanical energy conservation in frictionless systems.

Work done by constant and variable force fields.

LEARNING OBJECTIVES & KEY CONCEPTS

Problem solving: Step-by-step energy balance calculations.

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Chapter 3: Geometric Optics and Lenses

Snell-Descartes laws of refraction and reflection.

Thin converging and diverging lenses: focal length and optical centers.

Image formation for real and virtual objects.

LEARNING OBJECTIVES & KEY CONCEPTS

Laboratory method: Aligning benches and screen measurement.

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Chapter 4: Chemical Reaction Kinetics

Rate of reaction and collision theory.

Factors influencing reaction velocity: concentration, temperature, catalysts.

Spectrophotometric monitoring of chemical transformations.

LEARNING OBJECTIVES & KEY CONCEPTS

Experimental safety: Personal protective equipment protocols.